

Project Charter

Project Title: FaceRoll – Face Recognition Attendance System

Team Name: Cowboys

Identification Section

Project Name: FaceRoll

Team Name: Cowboys

Team Lead: Gabriel Oniyitan

Team Members: Austin Woodruff, Moses Ojo, Cade Powell, Isaiah Bailey

Course: CSCE 4905.501 - Information Technology Capstone I

Project Duration: 01/12/2026-05/08/2026

Overview of the Project

FaceRoll is a facial-recognition attendance system that consist of a student mobile application, Firebase services, and locally hosted facial-recognition backend. FaceRoll was created to automate classroom attendance in higher education settings. Having a teacher or professor call roll is no longer needed because FaceRoll records student attendance through an active facial-recognition check-in process during attendance sessions. FaceRoll synchronizes with course rosters and gains attendance information by integrating with Canvas's Learning Management System (LMS) through its API. With this, it will be able to track arrival times as well as absences. All facial recognition processing is done locally rather than on the cloud to reduce any privacy concerns.

Objectives

- Automate attendance tracking to reduce the time spent on calling roll in class
- Improve attendance accuracy and prevent proxy attendance from happening
- Seamlessly integrate with Canvas LMS
- Provide professors and teachers with a well-made interface for managing attendance

Scope

The scope of FaceRoll includes the design and development of a facial-recognition attendance system with a student mobile application, facial-recognition backend, Firebase-based authentication and storage. This system utilizes facial recognition technology to support course roster synchronization with Canvas LMS. The

project will focus on automatic attendance logging. Securing the storage for the data, as well as creating an easy-to-use interface for professors and teachers to review the attendance data.

Major Milestones

- Project requirements finalized: February 1, 2026
- Develop user stories: February 8, 2026
- SRS documentation: February 15, 2026
- Establish project plan in Trello: February 15, 2026

Major Deliverables

- Functional student mobile app
- Facial recognition with local processing
- Canvas LMS API integration for attendance updates
- Dashboard with attendance management and reporting
- Secure local storage system for facial data
- Finalized project documentation (user guide)
- Firebase-backed attendance storage
- Project presentation and demonstration

Assumptions & Constraints

Assumptions:

- The school/institution allows the use of face recognition for attendance
- Students will opt-in / provide consent
- Instructors / authorized staff will have access to a camera with adequate quality
- Lighting and camera placement will be generally sufficient for most face capture sessions.
- Canvas access is available through canvas API, the institution can provide any API access
- The system will be used in a web browser by instructors
- The team can use an existing face recognition model/library rather than training a model from scratch
- Student rosters and course sections are maintained in Canvas and are accurate enough to sync

Constraints:

- The project must be fully designed, implemented, tested, and documented within the defined project timeline of January 2026 through December 2026.

- Integration capabilities are constrained by Canvas LMS API rate limits, permissions, and documentation
- The system is intended for use in controlled classroom environments and is not required to support large-scale, campus-wide, or multi-institution deployment.
- Facial recognition accuracy may be affected by lighting, camera placement, and student movement. The system is not required to guarantee 100% accuracy under all conditions.
- The system must comply with FERPA, university data governance policies, and all applicable privacy and ethical standards related to biometric and student data.

Business Need / Opportunity

Manual attendance is inefficient and prone to errors. FaceRoll addresses this problem by providing an automated, accurate, and privacy-conscious solution that integrates with existing academic infrastructures like Canvas.

Risks

- Privacy and ethical concerns with data
- Inaccurate recognition / Model bias
- API limitations and integration
- Lack of student consent
- Performance issues when processing large classes
- System reliability
- Institutional policy restrictions on biometric technologies
- Training and Onboarding Challenges
- Low adoption by instructors and students

Stakeholders

- Project Developers: Austin Woodruff, Cade Powell, Isaiah Bailey, Gabriel Oniyitan
- Users: Instructors, teaching assistants, university departments
- Project sponsor: Professor Rabah

Approval

Project Leader: Gabriel Oniyitan Date: 1/27/26

Sponsor: _____ Date: _____